

The Next Test

A game of experimental design and identification

Companion to the standalone HTML game

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Abstract. The Next Test asks players to choose studies that establish a causal quantity, or establish that the available studies cannot determine it. Each case supplies a finite, explicit set of possible explanations, a fixed hidden explanation, and a menu of studies with exact predicted results. A visual diagram groups explanations by the results they predict. Observations eliminate incompatible explanations. Players submit either a numerical answer supported by every surviving explanation, or an argument for non-identification. The game computes shortest adaptive strategies and distinguishes identification of a target from identification of the entire explanation. Nine cases address selection, heterogeneous effects, useful covariates, individual treatment effects and spillovers. This note specifies the mechanics, the meaning of non-identification, and the two distinct experiment-count benchmarks.

1 The player’s task

A case begins with a concrete policy question: for example, how much a grant changes average income, or what percentage of people benefit. The starting facts are followed by explanation cards. Each card states a complete possible situation and its answer to the question. The cards are an *exhaustive model class*, not merely a few illustrative guesses. One is selected at the beginning of a round and remains true throughout it.

The player selects a study, such as randomising a grant, testing a programme in one district, or retaining district labels in the results. Before running it, the player sees a branching diagram: explanation letters on the left connect to their predicted results on the right. Explanations predicting the same result share a branch. This preview is free. Running the study costs one experiment and reveals the result implied by the hidden explanation. Incompatible cards are then marked as ruled out.

The player can open *Give an answer* at any time. For a numerical conclusion, the player selects the number directly. A correct guess is insufficient: all remaining cards must imply it. For a non-identification conclusion, the player compares two surviving explanations that disagree about the target but agree on every available study. An additional check prevents this argument from ending a case prematurely when another still-possible situation could yield an exact answer.

There is no time limit or hard experiment budget. The objective is a justified conclusion reached economically. After a successful submission, the game displays the number of experiments used, a shortest guaranteed strategy, and an explanation of the economic issue. The full decision tree is available for inspection. Study histories remain in a notebook; rereading a result costs nothing.

What counts as an experiment? The counter treats each information-gathering action as one unit, including observational surveys and diagnostic measurements. The menu states which actions involve random assignment. All results are exact population summaries: the game isolates identification and study choice from sampling uncertainty.

2 Information, identification and impossibility

Let $\mathcal{H}$ be the finite set of explanations and $h^* \in \mathcal{H}$ the hidden one. The target is a scalar $\theta(h)$. Each available study $e \in \mathcal{E}$ has a known deterministic observation function $f_e : \mathcal{H} \rightarrow \mathcal{Y}_e$. Results may be means, differences, distributions or a diagnostic label. A study uses a fresh comparable population; its result does not change the hidden explanation or the results of other studies.

After studies $e_1, \dots, e_t$ and observations $y_1, \dots, y_t$, the surviving explanations are

$$S_t = \{h \in \mathcal{H} : f_{e_j}(h) = y_j \text{ for every } j \leq t\}.$$

The identified set is $\Theta(S_t) = \{\theta(h) : h \in S_t\}$. The requested number is point identified precisely when $\Theta(S_t)$ is a singleton. Several explanations can survive while the target is already identified. Conversely, a hidden explanation may happen to imply the player's proposed number without the available evidence supporting that number uniquely.

Define observational equivalence under the *entire available menu* by

$$h \sim h' \iff f_e(h) = f_e(h') \text{ for every } e \in \mathcal{E}.$$

Write $[h]$ for the resulting equivalence class. Since survivors arise by conditioning on these same observation functions, every class containing a survivor is contained in the surviving set.

Proposition 1. *A strategy guarantees an exact numerical answer for every hidden explanation if and only if θ is constant on every observational-equivalence class.*

Proof. If equivalent explanations disagree about θ , they produce identical histories under any adaptive strategy. The strategy therefore cannot justify different numerical answers in the two situations. Conversely, running every available study reveals the complete observation signature and hence the equivalence class. Constancy of θ on that class then gives an exact answer. $\square$

For a particular hidden situation h^* , an exact number is attainable if and only if θ is constant on $[h^*]$. This local statement matters: a case may contain both situations that can be identified and situations that cannot.

A justified stopping rule. A numerical submission is accepted when $|\Theta(S)| = 1$. A non-identification submission is accepted only when every equivalence class still possible contains more than one target value. Thus no still-possible hidden situation can eventually yield a uniquely supported number. The player must also exhibit a pair of surviving, equivalent explanations with different targets. In all nine authored cases, a terminal non-identification set consists of one equivalence class, so this pair directly demonstrates the unresolved disagreement.

A pair of equivalent explanations with different targets proves that *no strategy guarantees a number across the entire model class*. By itself it does not prove that the actual situation cannot be identified. If another surviving class has a unique target, further study may reveal that class. The game checks this distinction rather than rewarding an early impossibility claim.

3 The minimum number of experiments

The benchmark concerns an adaptive strategy, not a lucky sequence of results. After each observation, the next study may depend on what was learned. For a surviving set S , a study partitions it into nonempty result groups

$$\mathcal{P}_e(S) = \{\{h \in S : f_e(h) = y\} : y \in f_e(S)\}.$$

Every study costs one unit. A study constant on S adds no information and cannot be part of a shortest strategy.

Guaranteeing a number. Let $D_\theta(S)$ be the smallest worst-case number of further studies that guarantees point identification. The exact recurrence is

$$D_\theta(S) = \begin{cases} 0, & |\Theta(S)| = 1, \\ 1 + \min_{e: |\mathcal{P}_e(S)| > 1} \max_{G \in \mathcal{P}_e(S)} D_\theta(G), & \text{otherwise.} \end{cases}$$

An empty minimum is $+\infty$, and an infinite branch makes that study's worst-case cost infinite. Thus $D_\theta(\mathcal{H}) = \infty$ whenever some observational class contains conflicting target values.

Guaranteeing a justified conclusion. The game also needs a finite benchmark for cases in which a number cannot always be learned. Let $T(S)$ hold when either the target is already point identified, or every observational class in S has multiple target values. Define

$$D_C(S) = \begin{cases} 0, & T(S), \\ 1 + \min_{e: |\mathcal{P}_e(S)| > 1} \max_{G \in \mathcal{P}_e(S)} D_C(G), & \text{otherwise.} \end{cases}$$

This is the fewest studies that guarantee reaching either a supported number or a supported non-identification conclusion. It is always finite: the full set of studies reveals the observational class. The game's *Shortest guaranteed plan* displays $D_C(\mathcal{H})$ and explicitly says when no strategy guarantees a numerical answer.

Proposition 2. *The two recurrences give the corresponding optimal worst-case experiment counts.*

Proof. Every nonterminal strategy must choose a first study. Nature's possible result leaves one of its groups G , where the appropriate continuation requires at least the optimal cost for G . This gives the lower bound. Choosing a minimising first study and then a minimising continuation attains it. Recursion terminates because every study considered splits S into strictly smaller sets. $\square$

The engine memoises these recurrences and constructs an optimal decision tree. A player's realised experiment count can be lower than the initial worst-case benchmark. For example, the first district tested may immediately reveal the only responding district and its effect. Feedback describes this as an early resolution on that outcome, not as beating a mathematical lower bound.

For each chosen study the game also checks whether $1 + \max_G D_C(G)$ equals the current $D_C(S)$. This evaluates the choice using the information available *before* its result. An uninformative study, or a study taken after a conclusion was already justified, does not pass that efficiency check. No prior probabilities over explanations enter the benchmark.

4 The nine cases

Table 1 reports the exact initial benchmarks. An infinity in the numerical column is a genuine identification failure under the available menu; it is not an exhausted budget. Zero in the conclusion column means that the starting facts and predictions already establish the limitation.

	Case	Models	D_θ	D_C
1	Who signed up?	4	1	1
2	Two districts	4	2	2
3	Enough to answer	4	1	1
4	Where to look next	4	2	2
5	Who actually benefits?	4	∞	0
6	Keep the district labels	6	1	1
7	Let the result decide	4	∞	1
8	Beyond the recipients	4	2	2
9	Three independent pilots	8	3	3

Table 1: Minimum worst-case numbers of studies.

Selection and heterogeneity (cases 1–4). In case 1, half the population voluntarily receives the grant. Recipients earn 8 and non-recipients 4. A constant causal effect $t \in \{-2, 0, 2, 4\}$ is consistent with these observations: volunteers’ untreated income is $8 - t$, and non-volunteers’ untreated income is 4. A representative randomised trial reveals t ; repeating the voluntary comparison always returns 4.

In case 2, district effects $n, s \in \{0, 4\}$ give a national gain $(n + s)/2$. Each available trial reveals one effect, so both districts must be tested. Case 3 instead has response functions $Y(d) = 4 + td + kd(1 - d)$, with $t \in \{2, 4\}$, $k \in \{0, 4\}$ and subsidy strength d . Full strength reveals the target t while leaving k unknown. In case 4, only North or South responds, by either 2 or 6. A positive district result identifies the national effect immediately; a zero result makes the other district the relevant next test. Two studies guarantee an answer.

Individual effects (cases 5–7). Case 5 has the same exact marginal distributions under every explanation, but different pairings of potential incomes. Every available study has the same prediction across explanations. Case 6 adds fixed district labels: untreated incomes are 0 throughout North and 2 throughout South. Labelled treated-income distributions identify the fraction who gain within each district, despite incomplete knowledge of individual matches. Case 7 mixes situations where marginal distributions suffice with situations where they do not; the example on the next page explains its stopping rule.

Spillovers and three pilots (cases 8–9). Case 8 specifies

$$Y_i(d_i, s) = 4 + \tau d_i + \gamma s, \quad \tau \in \{0, 2\}, \quad \gamma \in \{0, 4\}.$$

Only half-coverage pilots are permitted. A within-village contrast reveals τ ; a comparison of untreated people across randomly assigned half-coverage and zero-coverage villages reveals $\gamma/2$. Both are needed for the full-coverage target $\tau + \gamma$. Case 9 has three equal districts, each with effect 0 or 6. Every untested district leaves national effects differing by 2, proving that all three district trials are necessary.

5 Why the result can determine identifiability

Case 7 starts with an established average gain of 1 but no published income distributions. Its four explanations are given in Table 2. Each entry represents one quarter of the population; corresponding entries within an explanation concern the same people.

Explanation	Without programme	With programme	Benefiting
A	(0, 0, 0, 0)	(1, 1, 1, 1)	100%
B	(0, 1, 2, 3)	(4, 1, 2, 3)	25%
C	(0, 1, 2, 3)	(4, 3, 2, 1)	50%
D	(0, 1, 2, 3)	(1, 2, 3, 4)	100%

Table 2: Equal average effects, different identification possibilities.

A trial publishing the full marginal distributions has two possible results. If it finds only income 0 without the programme and only income 1 with it, A is the sole surviving explanation and the benefiting share is 100%. Otherwise B, C and D remain. They imply different shares, yet their predictions agree under every available study. Another trial cannot distinguish them.

Comparing B and C before doing any study already shows that no strategy guarantees a number across all four explanations. However, stopping there would miss the possibility that the trial identifies A. One informative study is therefore necessary and sufficient for a justified conclusion: $D_C = 1$, while $D_\theta = \infty$. This is why the interface can reject a premature impossibility argument even when the selected pair itself is observationally equivalent.

6 Scope, implementation and teaching purpose

Identification is conditional on the stated model class and study menu. The cards enumerate the possibilities; the rules state restrictions such as constant district effects or proportional spillovers. An assumption cannot be added as if it were an observation. Extrapolating to full coverage in case 8 depends on its stated response equation.

Exact population summaries remove sampling error. Repeated trials give the same result here because their population observations are already known; larger samples can still be valuable in applied work. No study observes both potential outcomes for one person under identical conditions. Individual effects remain unidentified when different couplings produce identical observations. Empirical problems generally admit many more explanations than these teaching cases.

The standalone HTML contains the cases, inference engine, shortest-strategy solver and interface, with no external runtime dependencies. Completed-case markers are stored locally where browser storage is available. There is no timer or network reporting.

Verification. Independent calculations reconstruct all 42 explanations from their economic primitives, enumerate all experimental histories and reproduce the nine benchmarks. A DOM simulation tests the interface handlers through every hidden situation, including rejection of unsupported conclusions. These checks do not substitute for visual browser testing of the rebuilt HTML.

The interface handles bookkeeping and arithmetic. The challenge is to choose informative studies and recognise when the available evidence is sufficient, or cannot answer the question.